

Stochastic and Risk-Aware Framework for Adaptive Healthcare Referral

Matteo Gomez, Department of Mathematical Sciences

Mentor: Dr. Satyaki Roy, Dept. Math; Dr. Jennifer Bail, College of Nursing

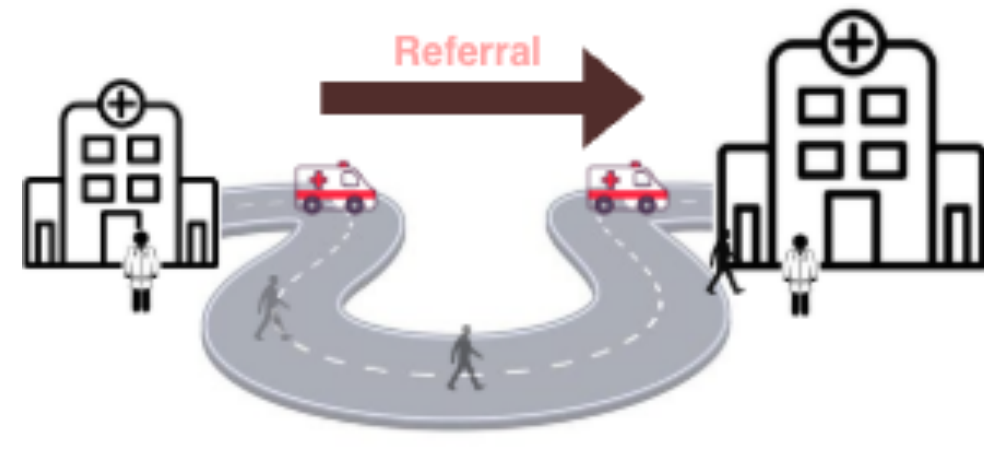
Overview

Motivation

- Hospital-acquired infections (HAIs)** are major healthcare burden, ~518,000 infections reported in U.S. acute care hospitals in 2023 [1].
- Patient transfers** and **referrals** can influence infection exposure [2, 3].

Broad Goals

- Model patient transfers through healthcare network **stochastically**
- Incorporate **infection risk** and **patient privacy** into **clinical decision-making**



Objective and Method

Objective

- Express** quality of care for given patient movement as **reward values**
- Optimize referral decisions by **balancing clinical compatibility, infection risk, and privacy**
- Validate** referrals with regression analysis to reward values



Figure 1: 1000+ node hospital network

Patient-Hospital Transfer Matrix		
Patient	Hospital 1	Hospital 2
1		0.65
2	0.8	
3	0.3	
4		0.5

Table 1: Sparse transfer matrix gives a partial picture of patient rewards

System Model

Healthcare Referral Network (HRN)

- Directed graph $G = (V, E)$
- $u \in V$ represents a healthcare center, arc $(u, v) \in E$ denotes patient referrals from u to v [4, 5].
- Hospitals have **contagion potential (CP)** — measure of infection risk

Patient-Hospital Transfer Matrix

- Captures the *observed* health outcome of referring a patient to a specific hospital following **importance sampling** to tackle **sparsity**
- Other **reward values** for patient movement are left unobserved
- Referrals follow overlaying **HRN**: no referrals over non-existent arcs

Optimization Process

Parameters

Patient feature matrix X , Number of patients n , Number of patient transfers m , Number of hospitals K , Regularization parameter for entropy λ , Contagion Potential CP

Infer

Patient Tracker $T = \{T_1, T_2, \dots, T_m\}$ is built to monitor movement within the network, with $T_i = (pat_i, source_i, dest_i)$. Patient i contains feature data $feat_i$.

Preprocessing

Patient and hospital features, then referral events are tracked and converted into rewards. For patient i ,

$$R_i = H_{sim}(T_{pat_i}, T_{dest_i}) / CP(dest_i)$$

Optimization

Weight values are learned to estimate the patient reward observed from the patient transfer matrix, then optimized.

$$\min_W - \left(\underbrace{\frac{K}{|T|} \sum_{t \in T} R_t W_{feat_t, dest_t}}_{\text{reward / data-fitting term}} + \underbrace{\lambda \sum_{i=1}^{|X|} \sum_{k=1}^K W_{X_i, k} \log(W_{X_i, k} + \epsilon)}_{\text{entropy regularization}} \right)$$

Results

Settings Python-based implementation [6, 7, 8] on HRN dataset [9] ($n = 50$, $K = 7$, $m = 100$)

Figure 2: Strong positive correlation between empirical reward and learned weight values (100 iterations).

Figure 3: Increased λ keeps trend but adds uncertainty in learned weight values (100 iterations).

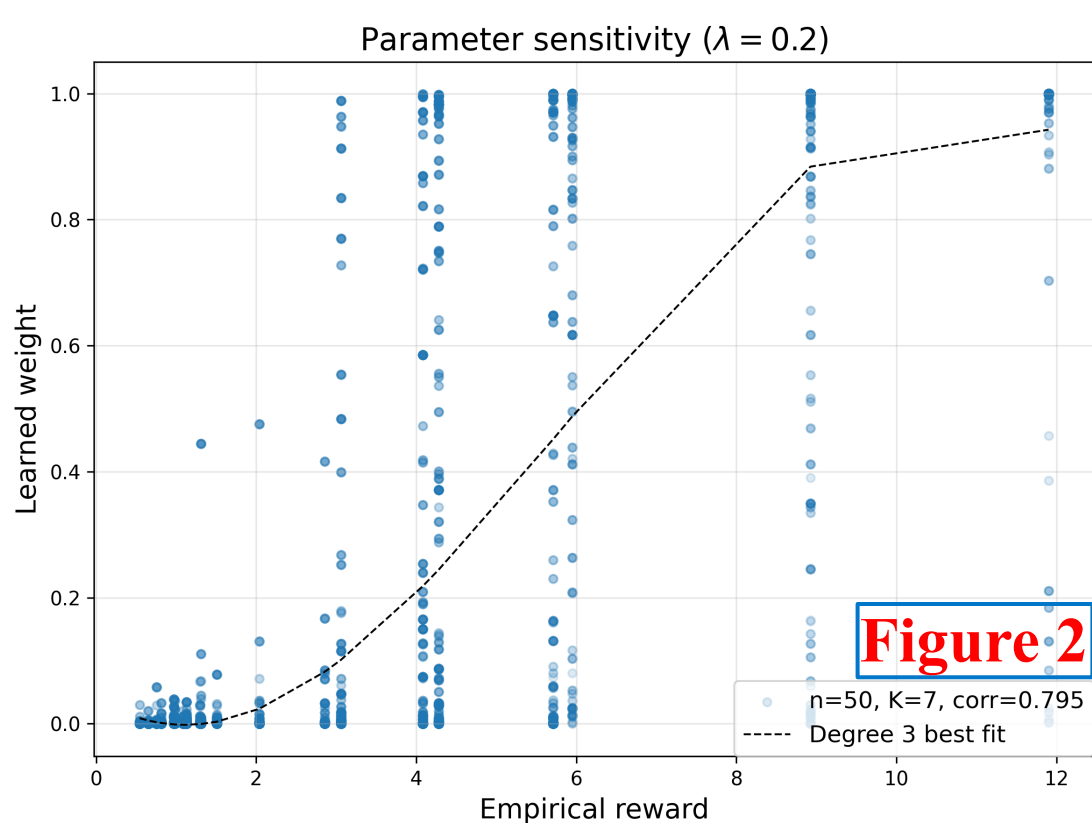


Figure 2

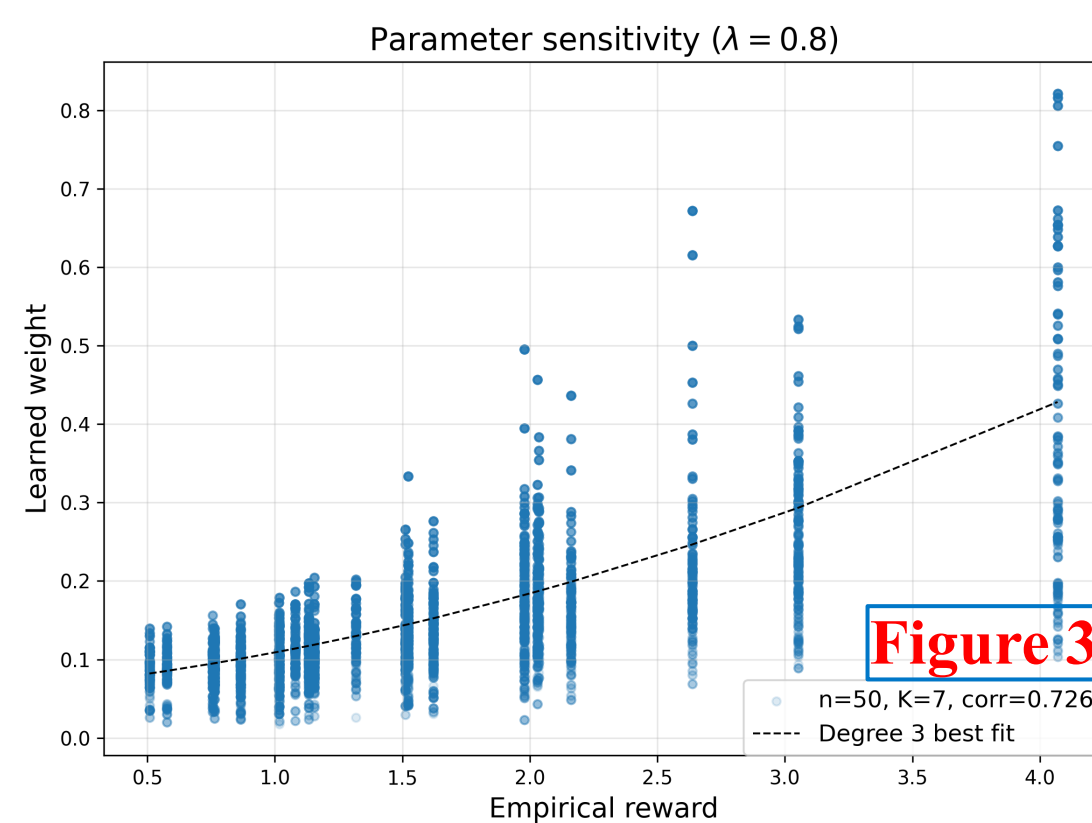


Figure 3

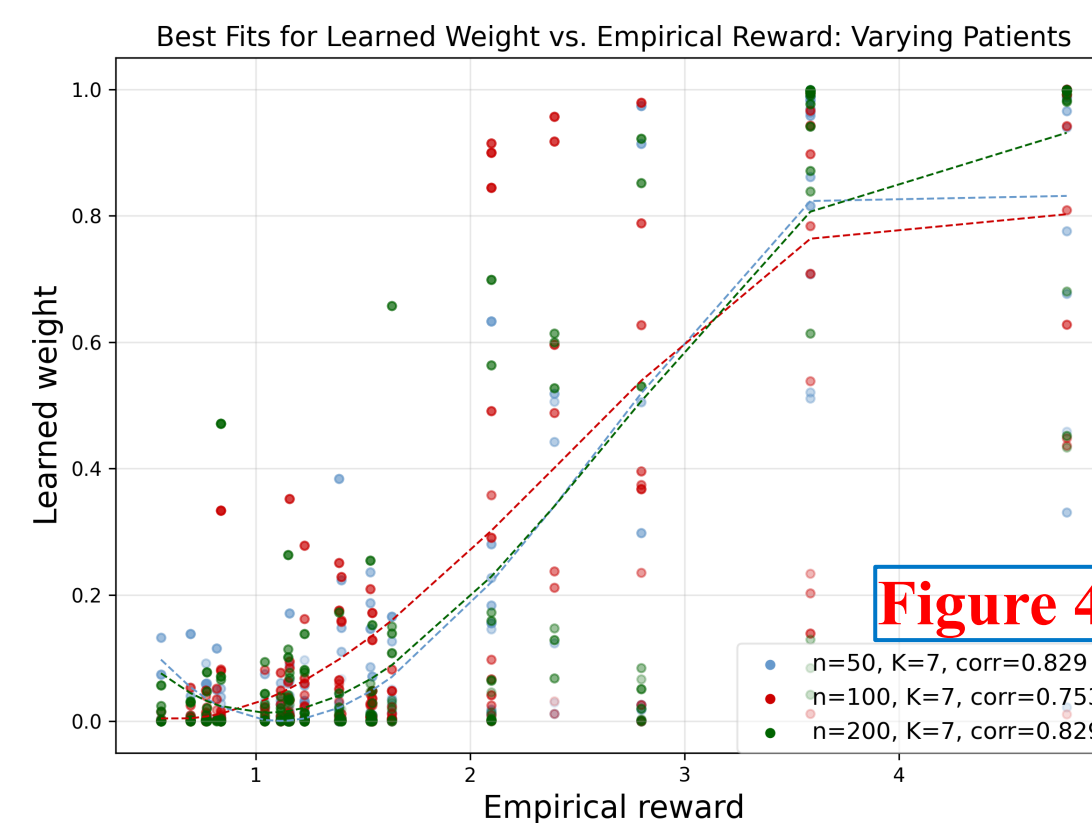


Figure 4

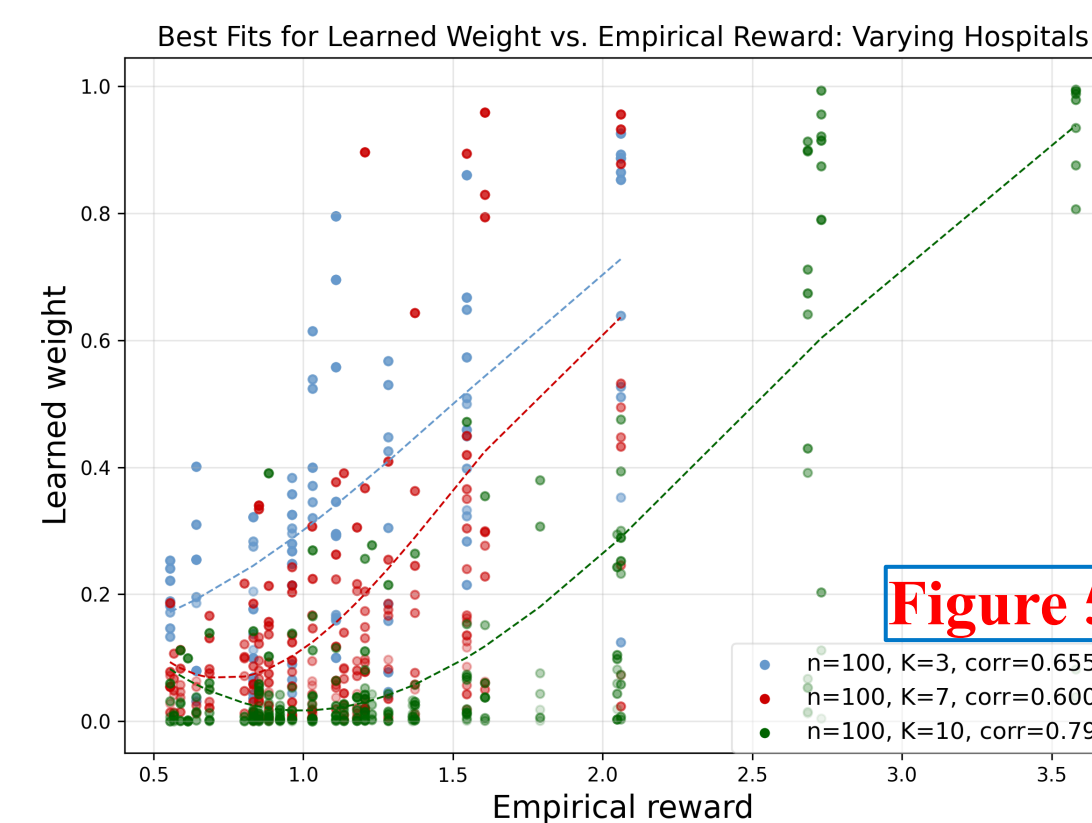


Figure 5

Figure 4: Varying patient counts (10 iterations) keeps positive trend in correlation between empirical reward and learned weights.

Figure 5: Varying hospital counts is more volatile because of a bigger search space, but keeps strong positive correlation.

Conclusions

- Correlation analysis suggest referrals meet **clinical needs**, expose patients to **low risk**, and preserve **patient privacy**.
- Results hold for varying numbers of hospitals and patients.
- Risk-aware framework shows promise in real, evolving clinical settings**

Ongoing

- Validate on large-scale HRN datasets
- Consider different disease groups

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